

VolStrat

A volatility-regime allocator: one daily risk-appetite score drives a tactical portfolio, backtested in the browser

app.kevinlhc.com/volstrat · October 2026

Goal: cut the big drawdowns, keep most of the upside

Worst peak-to-trough loss, 2007 to Oct 2026

−23.8%

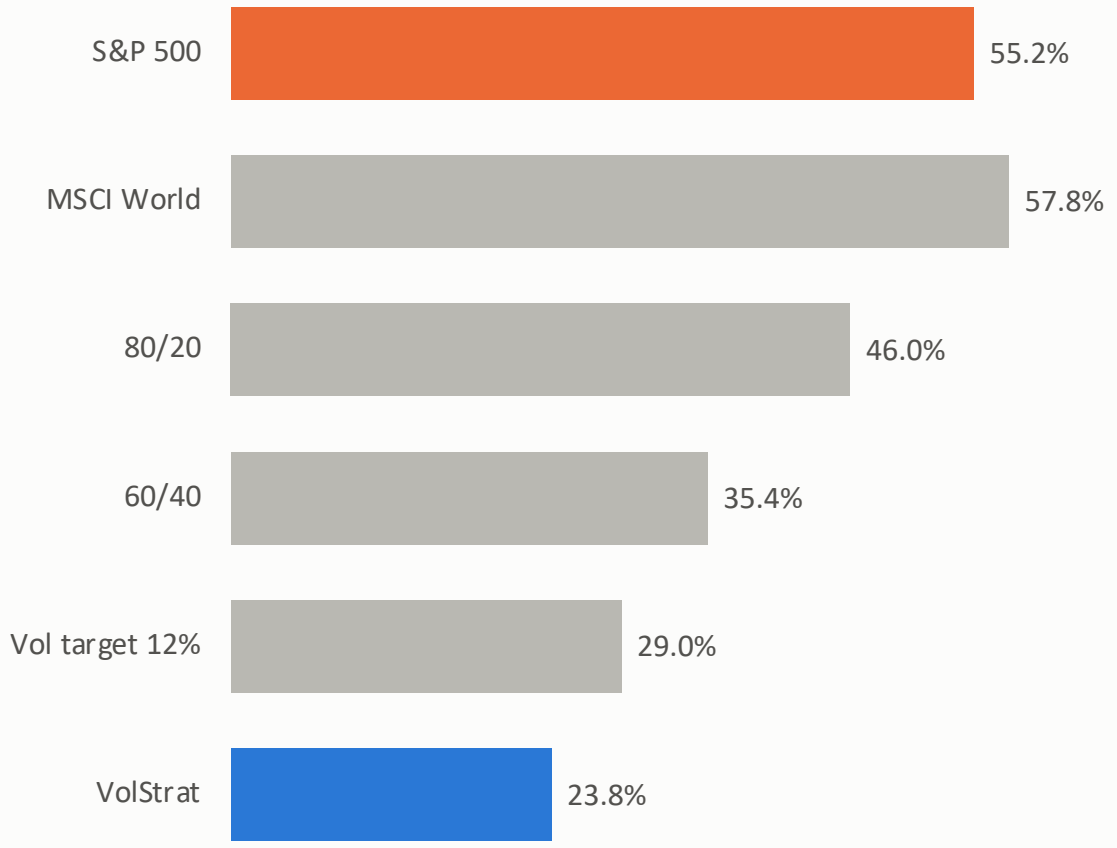
VolStrat (Growth profile)

−55.2%

S&P 500 buy and hold

CAGR 9.7% vs 10.9% for the S&P 500 at about two-thirds of its volatility (12.7% vs 19.5%)

Maximum drawdown, % (same period, same costs)





How it works

Data → score → regime → portfolio

From public market data to a daily allocation



1 Collect

Cboe indices, CFE VIX futures, put/call, Yahoo ETFs, US Treasury curve, FRED credit



2 Score

9 components ranked against their own trailing 3 years, then averaged into 0–100



3 Classify

Cut-offs 20 / 40 / 60 / 80 with a 2.5-point buffer give five regimes



4 Allocate

Each regime holds its own portfolio, traded at the next close



5 Backtest

Stats vs six benchmarks, recomputed in the browser on every change

No look-ahead by construction: every transform uses trailing windows only, each input carries its publication lag, and a test suite checks that future data never changes past decisions.

Nine signals, each read as a percentile of its own history

Component	What it measures	Weight
VIX level	Implied volatility	1
VIX / VIX3M	Term-structure inversion	1
VX futures slope	Backwardation, CM30 / CM60	1
Variance risk premium	VIX minus 21-day realised vol	1
Credit	HY vs Treasuries, 3-month return	1
Trend	S&P 500 vs 200-day average	1
VVIX	Volatility of volatility	0.5
Equity put/call	10-day average	0.5
Cboe SKEW	Tail-hedge demand (weak signal)	0.25

Trailing ranks

Each raw value is ranked among the last 756 trading days (≈3 years), so 100 is the most risk-on reading.

Weighted average

Missing inputs drop out and the remaining weights rescale. A 5-day EMA smooths the result.

All editable

Weights, windows, directions and 15 more optional signals can be changed in the Backtest panel.

Five regimes, each with its own portfolio

Crisis	Risk-Off	Neutral	Risk-On	Strong Risk-On
Score below 20	Score 20 – 40	Score 40 – 60	Score 60 – 80	Score above 80
100% T-bills	50% Treasuries (IEF) 25% gold · 25% utilities	80% S&P 500 20% Treasuries	75% S&P 500 25% SSO (2×) = 125% equity	100% SSO (2× S&P 500)

Share of trading days in each regime, 2007 – 2026



A regime only changes once the score crosses a cut-off by more than 2.5 points. Moves toward risk-off are never delayed.

The Dashboard: today's reading

Close of 22 Oct 2026

37.7

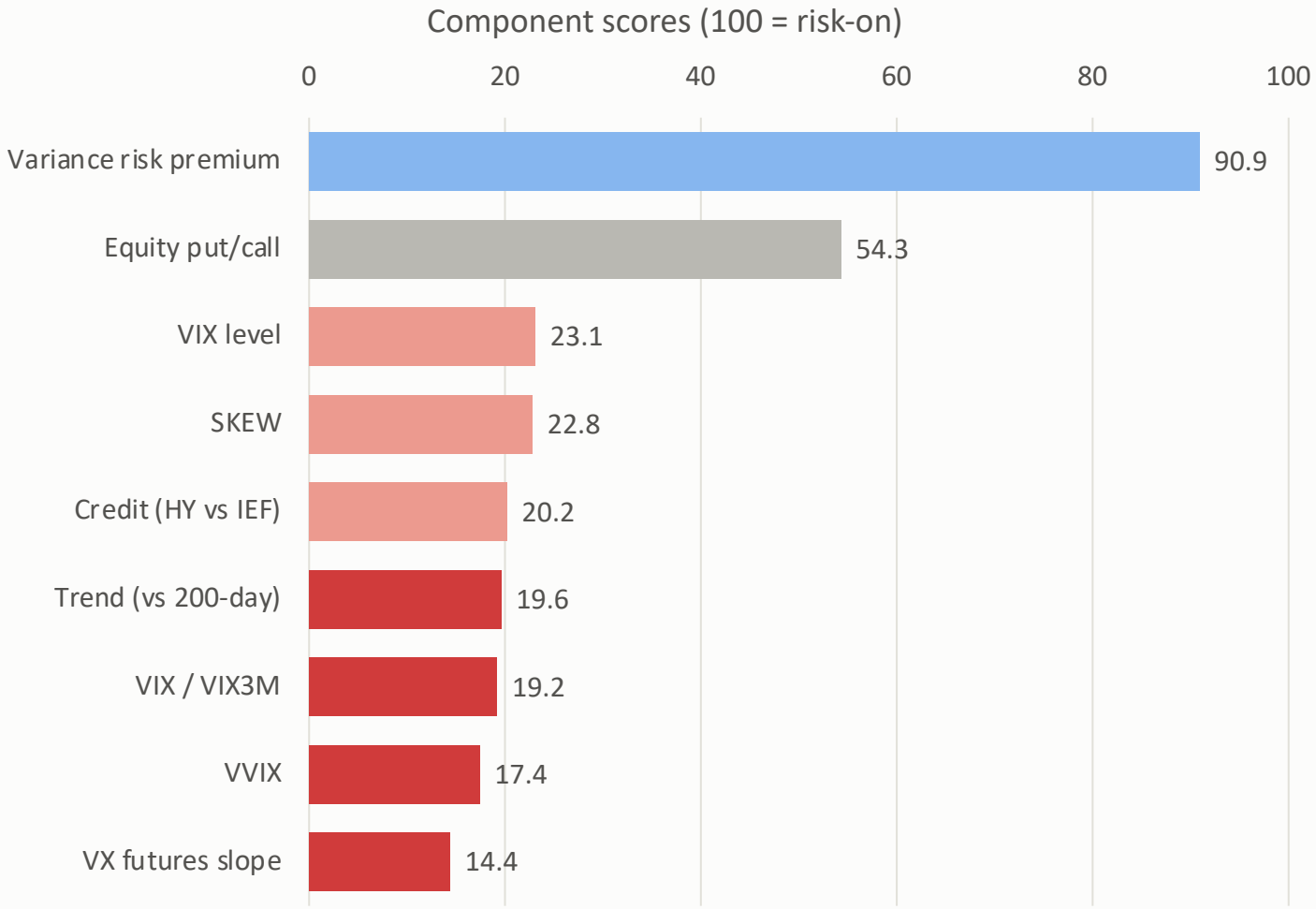
Risk-appetite score (0–100)

Neutral since 10 Sep 2026

Target allocation

80% S&P 500 · 20% Treasuries

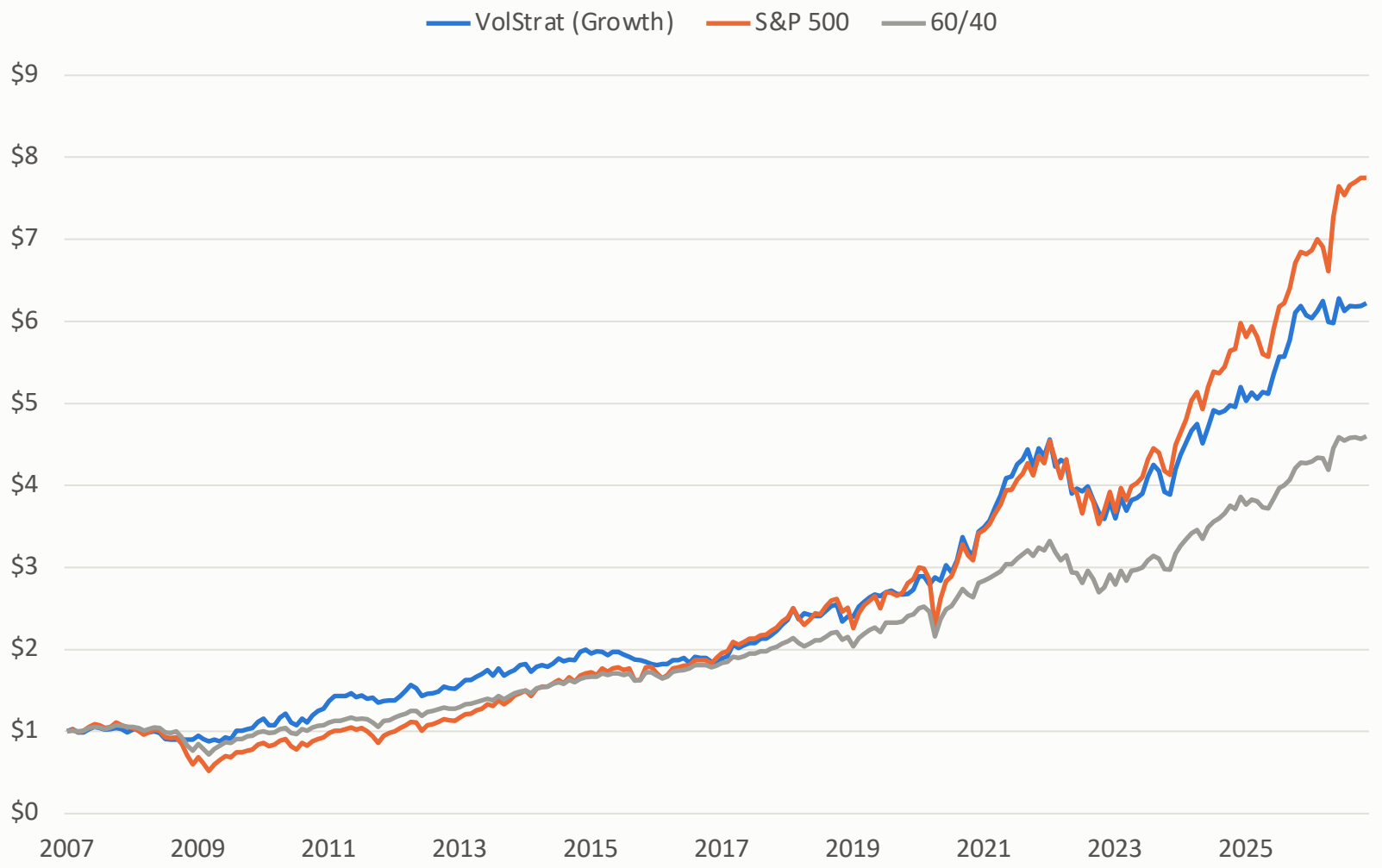
Traded at the next close



What the backtest shows

Jan 2007 – Oct 2026 · next-close execution · 5 bp per side

Growth of $1: most of the index's gain, far shallower dips



$6.22

VolStrat, ending value

$7.75

S&P 500

$4.60

60/40

Much smaller losses in 2008 and 2020, about the same in 2022, and a lag in sharp recoveries.

Best risk-adjusted return of the group

Portfolio	CAGR	Volatility	Sharpe	Max drawdown	Turnover / yr
VolStrat · Growth (default)	**9.7%**	**12.7%**	**0.66**	**−23.8%**	**6.6×**
VolStrat · Balanced (earlier default)	7.4%	10.0%	0.60	−21.8%	6.1×
S&P 500	10.9%	19.5%	0.54	−55.2%	–
MSCI World	8.4%	18.6%	0.44	−57.8%	–
80/20 S&P 500 / AGG	9.5%	15.5%	0.56	−46.0%	0.1×
60/40 S&P 500 / AGG	8.0%	11.7%	0.58	−35.4%	0.1×
Vol target 12% (S&P / T-bills)	8.6%	11.9%	0.61	−29.0%	2.3×



Highest Sharpe

0.66 vs 0.54 for the S&P 500 and 0.58 for 60/40



Costs matter

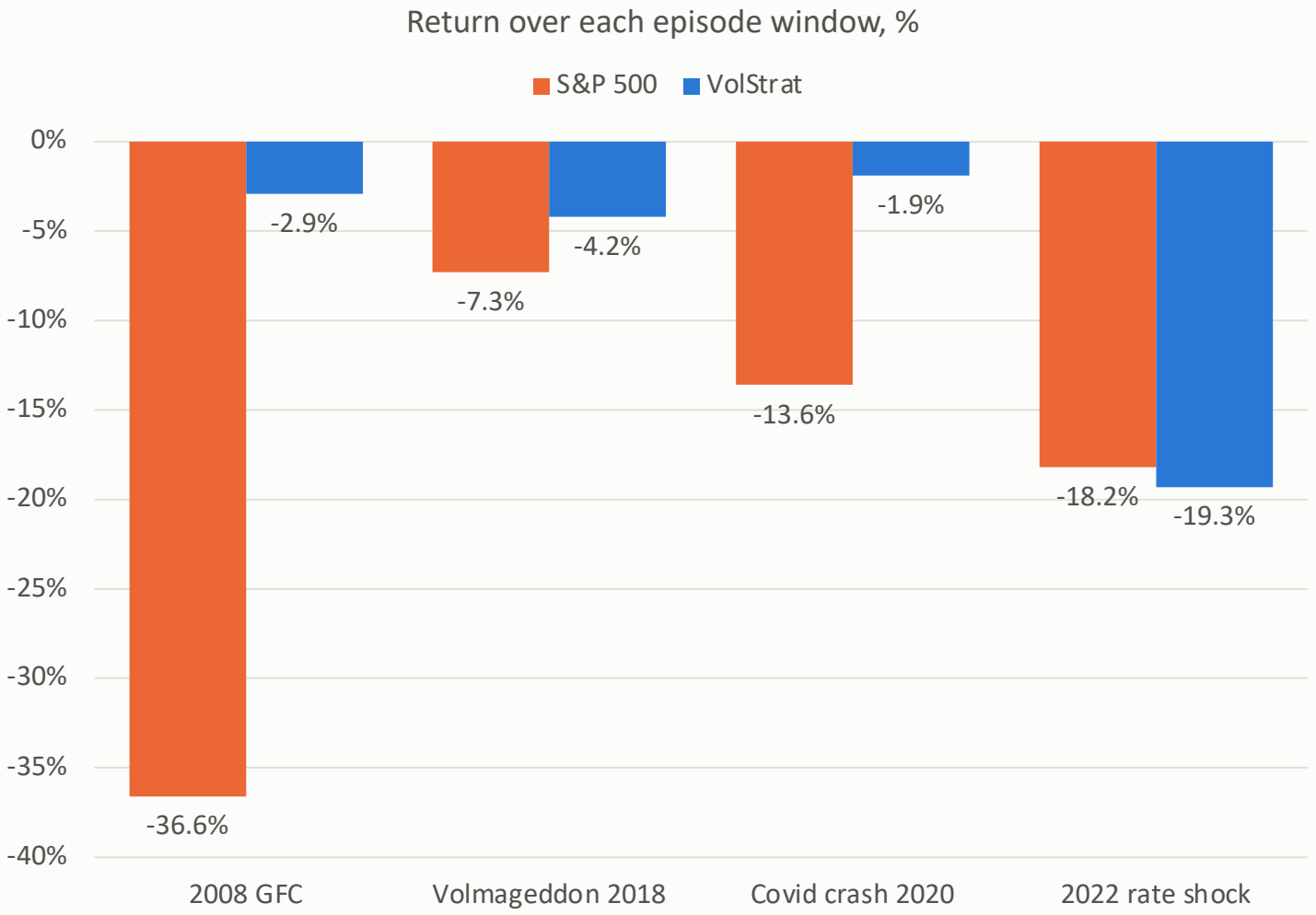
About 6.6× turnover a year, so execution cost is a first-order issue



Still trails the index

9.7% vs 10.9% CAGR, at less than half the drawdown

Volatility shocks absorbed; the 2022 rate shock was not



Fast signals lead

VIX/VIX3M inverted on 5 Feb 2018 while the trend filter barely moved. Mixing fast and slow signals is deliberate.

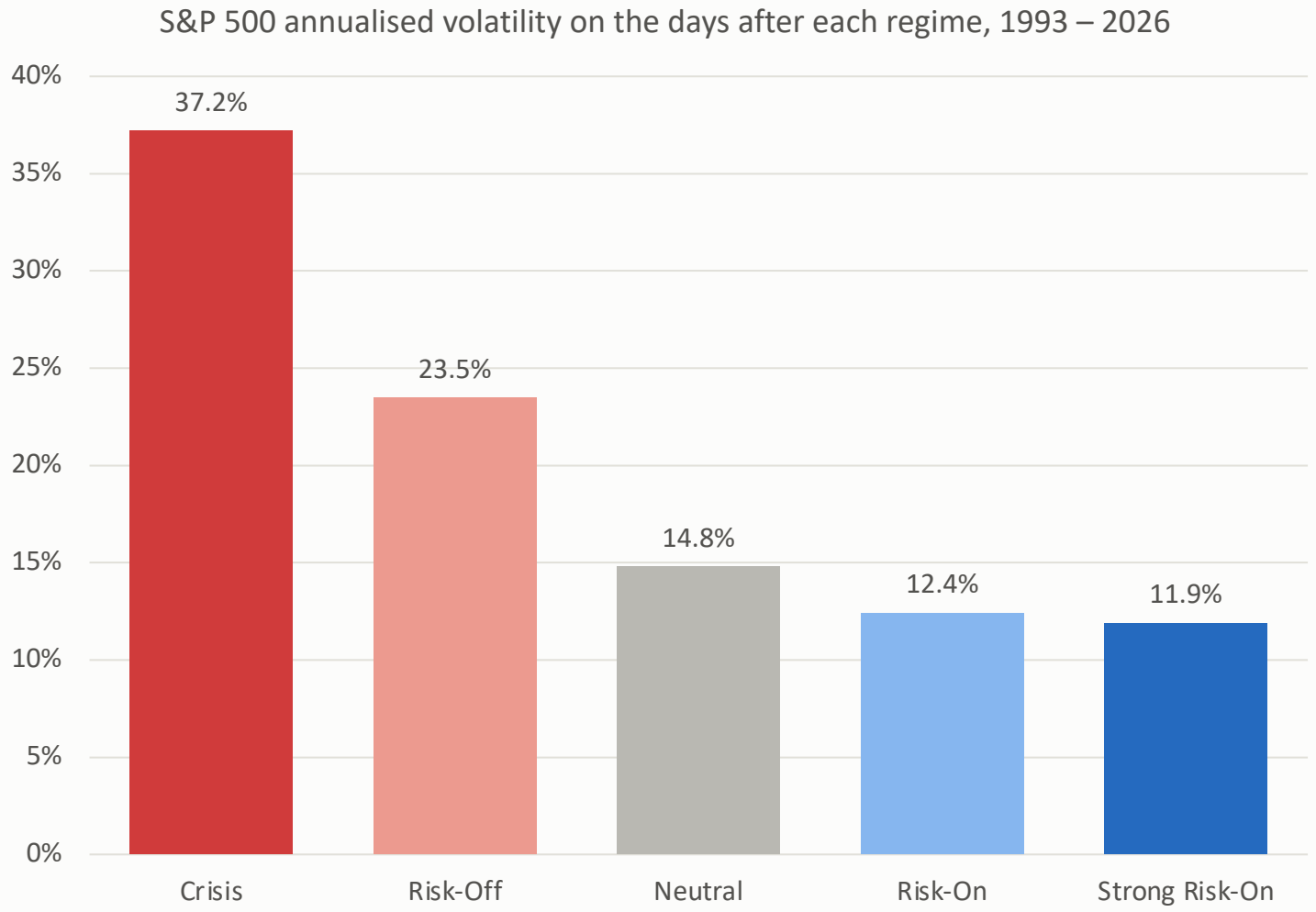
Right signal, wrong hedge

Stress was flagged by 21 Jan 2022, but in a rate shock Treasuries (−16.1%) and gold (−9.8%) fell too.

No hindsight fixes

A T-bill hedge would have helped 2022 but lowers the 2007–2026 Sharpe (0.53 vs 0.60), so it stays optional.

The score sorts risk, not return



3×

higher volatility after Crisis days than after calm days, and 3× the odds of a 10% loss within a quarter

0.66 – 0.72

Sharpe ratio of the S&P 500 after Crisis, Risk-Off, Neutral and Risk-On days: almost the same

−0.06

rank correlation of the score with next-month returns (t = −1.2): no return forecast



Guarding against overfitting

Every change is specified on 2007–2016 and judged once on 2017–2026

Most ideas fail out of sample, and the tool shows it

Candidate	Idea	Out-of-sample result	Verdict
Panic-rebound re-entry	Buy back after VIX/VIX3M inverts past 1.10 and normalises	Test Sharpe 0.75 → 0.85; better in 92% of resamples	**Adopted**
Higher exposure (Growth)	80/20 in Neutral, 125% equity in Risk-On	Return and drawdown rise together; Sharpe unchanged	**Risk dial**
FRED credit and stress indices	Baa−10y, NFCI, STLFSI as components	Gains came from 2008 only; worse in 70–96% after 2016	**Rejected**
Re-rank the composite	Spread the score evenly across regimes	Turnover doubles to 13×; worse in 94%	**Rejected**
One-sided VRP	Count the premium only on the stress side	Helped 2007–16, cost 0.31 Sharpe in 2017–26	**Rejected**
Broad recovery re-entry	Re-enter whenever volatility eases	Fired 70 times, often in 2008 bear rallies; better in 46%	**Rejected**

Built-in checks: sensitivity heatmaps, walk-forward selection (tuning scored 0.48 Sharpe vs 0.70 for fixed defaults), bootstrap intervals and a multiple-testing yardstick.

Seven pages, one engine, every setting in the URL

Dashboard

Today's score, regime, allocation, components and data status

Signals

24 components over the S&P 500 with regime shading; VIX futures curve

Backtest

Edit any parameter; stats vs six benchmarks, bootstrap and equal-risk checks

Diagnostics

Returns after each regime, information coefficient, leave-one-out

Optimise

Regime portfolios and signal weights, tested walk-forward

Compare

Saved parameter profiles side by side

Methodology

Plain-English documentation with live stress-episode tables

Alerts

RSS feed and a GitHub issue whenever the regime changes

Runs itself every trading day, with an honest live record

Hourly trigger

A Cloudflare Worker dispatches the data job every hour at :20; it commits only when data changed

Completeness gate

A date is published only once every input has posted, so the score never uses a partial day

Data guards

Range checks, cross-source reconciliation of the past week; a blank never overwrites a value

Frozen track records

Parameters frozen, decisions appended before the trade, never edited or backfilled

Live, out-of-sample track records

v1 · Balanced

Frozen 22 Oct 2026
First decision: 21 Oct 2026 close
Neutral · 60% S&P 500 / 40% IEF

v2 · Growth

Frozen 23 Oct 2026
First decision: 22 Oct 2026 close
Neutral · 80% S&P 500 / 20% IEF

The two profiles now run side by side out of sample.



Takeaways

- A transparent, browser-based lab for volatility-regime allocation
- Drawdowns cut by more than half (−23.8% vs −55.2%), Sharpe 0.66 vs 0.54
- The edge is risk control, not return forecasting; it trails the index's CAGR
- Live track records will be the real test

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